

Python Module End Assignment

Step 1: Preloaded Ticket:

```
import string

# Step 1: Preloaded Tickets
ticket_data = {
    'Ticket_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Customer_Name': ['Ravi', 'Meera', 'Sam', 'Anu', 'Rakesh',
                     'Divya', 'Arjun', 'Kiran', 'Leela', 'Nisha'],
    'Issue_Description': [
        'Internet not working!!! ', 'slow response, very poor service ',
        'GREAT support! issue resolved.', ' okay... need help ',
        'not BAD but slow', 'Excellent guidance, Very Helpful!',
        'good support and good behaviour!', 'Poor handling of technical issue',
        'Satisfied. Could be better.', 'Good service... quick response.'
    ],
    'Priority': ['High', 'Low', 'High', 'Medium', 'Low',
                'High', 'Medium', 'High', 'Low', 'Medium']
}

def show_data(data):
    print("{")
    for key, values in data.items():
        print(f"  '{key}': [")
        for v in values:
```

```
        print(f"    {v!r},")
        print("  ],")
    print("}")

print("=== INITIAL TICKET DATA ===")
show_data(ticket_data)
```

Step 2: Add more Tickets:

```
# Step 2: Add More Tickets
while True:
    try:
        n = int(input("\nHow many new tickets do you want to add? "))
        if n >= 0:
            break
        print("Enter 0 or a positive number.")
    except ValueError:
        print("Please enter a valid number.")

next_no = max(ticket_data['Ticket_No']) + 1  # starts from 11

for _ in range(n):
    name = input("\nCustomer Name: ").strip()
    issue = input("Issue Description: ")
    while True:
        priority = input("Priority (High/Medium/Low): ").strip().capitalize()
        if priority in ('High', 'Medium', 'Low'):
            break
        print("Invalid priority! Enter only High, Medium or Low.")

    ticket_data['Ticket_No'].append(next_no)
    ticket_data['Customer_Name'].append(name)
    ticket_data['Issue_Description'].append(issue)
```

```
ticket_data['Ticket_No'].append(next_no)
ticket_data['Customer_Name'].append(name)
ticket_data['Issue_Description'].append(issue)
ticket_data['Priority'].append(priority)
print(f"Ticket {next_no} added.")
next_no += 1
```

... Show hidden output

Step 3:Text Cleaning:

```
# Step 3: Text Cleaning
slang = {'ok': 'okay', 'pls': 'please', 'plz': 'please', 'thx': 'thanks',
        'u': 'you', 'ur': 'your', 'gr8': 'great', 'bcoz': 'because',
        'b4': 'before', 'asap': 'as soon as possible'}

def clean_text(text):
    text = text.lower()
    for p in ".!?-":
        text = text.replace(p, " ")
    words = text.split()
    words = [slang.get(w, w) for w in words]
    return " ".join(words)

ticket_data['Issue_Description'] = [clean_text(t) for t in ticket_data['Issue_Description']]
print(ticket_data["Issue_Description"])

... ['internet not working', 'slow response very poor service', 'great support issue resolved', 'okay need
```

Step 4:KeyWord-Based Insights:

```
# Step 4: Keyword-Based Insights
def count_tickets_with_word(word):
    word = word.lower()
    return sum(1 for desc in ticket_data['Issue_Description'] if word in desc.split())

print("\n=== KEYWORD INSIGHTS ===")
for w in ["poor", "good", "slow", "excellent"]:
    print(f'Number of tickets containing "{w}": {count_tickets_with_word(w)}')

... Show hidden output
```

Step 5: Final Summary & Insights:

```
# Step 5: Final Summary & Insights
# 1. Final cleaned data
print("\n=== FINAL CLEANED TICKET DATA ===")
show_data(ticket_data)

# 2. Priority analysis
print("\n=== PRIORITY ANALYSIS ===")
for level in ("High", "Medium", "Low"):
    print(f"Number of {level}-priority tickets: {ticket_data['Priority'].count(level)}")

# 3. Longest issue description (by word count)
word_counts = [len(d.split()) for d in ticket_data['Issue_Description']]
max_wc = max(word_counts)
print("\n=== LONGEST ISSUE DESCRIPTION ===")
for i, wc in enumerate(word_counts):
    if wc == max_wc: # prints all tickets if there is a tie
        print("Ticket number :", ticket_data['Ticket_No'][i])
        print("Customer name :", ticket_data['Customer_Name'][i])
        print("Cleaned issue :", ticket_data['Issue_Description'][i])
        print("Word count      :", wc, "\n")
```

```
# 4. Unique words
unique_words = set()
for desc in ticket_data['Issue_Description']:
    unique_words.update(desc.split())

print("=== UNIQUE WORDS ===")
print("Count of unique words:", len(unique_words))
print("Word list (sorted):", sorted(unique_words))
```